DAO2702X group project: what should a studio greenlight?

Outline: data, problem, visuals, models

Last touched 2026-10-19: first version, data downloaded, nothing coded yet.

> **The project in one line.** A studio has to decide *before filming* how much to spend, on what kind of film, with whom, and when to release it. We use about 3,600 films to show which of those choices raise expected box office, which raise or lower the risk of a flop, and what slate of films a studio should build given how much risk it can carry.
> **A first look at the data supports this** (The Numbers, 4,952 films 1980-2018): films over $120M lose money less often than films under $20M (40% vs 62% below break-even), and horror, not romance, is the cheap high-upside genre (median 2.75x budget, top 10% above 16x).

1. THE PROBLEM

How should a film studio choose budget, genre, source material, talent and release date so that it gets the best expected return for the risk it can take?

Most student movie projects predict IMDb score or gross from everything in the file. We only use what the studio controls or knows *before the film is made*. Ratings, Facebook likes, opening weekend and theatre counts all happen after release, so they are out of the model (they can still appear in descriptive charts, labelled as such).

We measure two outcomes, because a studio cares about the average return and about how often it loses money:

Outcome	Definition	Why
Gross multiple	worldwide gross / production budget	Return per dollar spent. Comparable across budget sizes
Flop	gross multiple < 2.5	Downside risk. 2.5x is the industry rule of thumb for break-even: the studio keeps about half the box office, and marketing costs roughly as much as half the production budget again

The 2.5x line is an assumption we state and justify in the report (the brief rewards this). We can show the main charts also hold at 1x and 3x.

Hypotheses to test, ranked by how good a chart they make:

1. Bigger budgets *lower* risk, not just raise revenue (early data says yes).
2. Romance is "free money" (early data says no: rom-com median 2.28x, 54% flop; horror beats it).
3. Releasing into a crowded month (many big films within 30 days) cuts the multiple.
4. A director or lead with a strong track record lowers the chance of a flop more than it raises the upside.
5. Big studios and small studios should rationally pick different slates, because they can carry different amounts of risk.

2. THE DATA

Sources, all downloaded

Everything is in `~/Documents/Programming/progforbiz/movie-studio/data/raw/`.

Rank	Dataset	Rows	What it adds	Role
1	`movie_data.csv` (course dataset, IMDb 5000 subset)	3,647 films, 1916-2016	budget, US gross, genres, director, 3 lead actors, content rating, runtime, country, inflation factor	**Base table**
2	The Numbers budgets (scraped by delafields, Dec 2019)	5,460 films, 1925-2019	**release date**, **worldwide gross**, domestic gross, story source, production method, creative type, production company, full cast list	Outcome + timing + source
3	TMDB 5000 (`tmdb_5000_movies.csv`, `_credits.csv`)	4,803 films	release date, **production companies**, keywords, worldwide revenue, full cast and crew	Fills gaps in 2, studio field
4	Lock5 HollywoodMovies (2007-2012 and 2012-2018)	~1,900 films	**lead studio (distributor)**, story type	Studio strategy
5	FRED CPI (`CPIAUCSL`)	monthly 1947-2026	consumer prices	One consistent inflation adjustment for all sources

3,579 of the 3,647 base films match at least one of TMDB or The Numbers on title and year (3,292 match both). So the joined table keeps 98% of the base.

Problems in the base data

1. **`gross` is US domestic only**, not worldwide. Worldwide comes from The Numbers or TMDB.
2. **Non-US budgets are in local currency.** *The Host* shows a budget of 12,215 (Korean won, millions). Use The Numbers' USD budget, or keep USA/UK films only.
3. **Facebook likes were collected around 2016**, years after most films came out. They measure fame earned partly *from* the film, so they leak the outcome. Out of the model.
4. 42 duplicated titles, 132 budgets under $1M, broken characters in names (`Zoí, Kravitz`). Names need fixing before matching people across films.
5. The Numbers scrape has shifted columns for some films (text like "All Time Domestic Box Office for…" sits in the genre field). Fix by taking genre from the base table and TMDB.

The joined table (one row per film)

Group	Columns	Studio controls it?
Identity	title, release_date, year	
Money	budget_real (2016 $), log_budget, budget_tier (<20M, 20-60M, 60-120M, 120M+)	Yes
Content	main_genre + genre dummies, story_source (original, book, comic, sequel, real events), production_method (live action, animation), content_rating, runtime	Yes
Talent	director and lead actor track record: number of prior films, median prior multiple, prior flop rate	Yes (who to hire)
Studio	production company / distributor, its prior film count, median multiple and flop rate	Partly
Timing	release month, holiday window, `competitors_30d`, `big_competitors_30d` (budget at least as large), `same_genre_30d`, `competitor_budget_sum_30d`	Yes
Outcomes	worldwide_gross_real, domestic share, gross multiple, flop flag	No: target
Post-release (charts only)	IMDb score, likes, opening weekend	No: never a model input

A person's or studio's track record uses only films released *before* the target film. Otherwise the model knows the future.

Competition counts use release dates from TMDB and The Numbers together (about 7,000 films). That list undercounts small releases, which we state as a limitation.

3. THE FOUR STAGES, AS IN LEC9

The course deck splits analytics into descriptive, predictive and diagnostic, and prescriptive. The report follows that order as one argument, each stage answering the question the last one raised, because the brief marks down reports written as separate tasks.

Stage	Question it answers here	Report pages
Descriptive	What do returns and risk look like across budgets, genres, timing?	2.5
Diagnostic	Which pre-production choices move the multiple and the flop risk, holding the rest fixed?	1.5
Predictive	For a new pitch, what range of outcomes should the studio expect?	1.5
Prescriptive	Given its risk appetite, what slate should a studio build?	1.5

Plus 1 page problem and data, 0.5 page limitations and recommendations.

4. VISUALS, RANKED

Visuals are 50% of the grade. One colour palette across all charts: one accent colour for the group we are talking about, grey for everything else, red only for the flop line. Every chart is a different type and answers one question.

Rank	Chart	Type	Stage	Question
1	Budget vs worldwide gross, log axes, with 1x and 2.5x break-even lines	Scatter	Descriptive	Does spending more pay? Opens the report
2	Gross multiple by budget tier	Violin or ridgeline	Descriptive	Does a bigger budget narrow the spread? (hypothesis 1)
3	Genre risk-return map: median multiple (y) vs flop rate (x), bubble = number of films	Bubble scatter	Descriptive	Is romance free money? Where does horror sit? (hypothesis 2)
4	Flop rate by genre x budget tier	Heatmap	Descriptive	Which combinations are safest?
5	Median multiple by number of big competitors within 30 days	Line with confidence band	Descriptive	Does a crowded release window hurt? (hypothesis 3)
6	Domestic vs international share by genre	Stacked bar	Descriptive	Which genres depend on foreign markets?
7	Director track record vs outcome	Box plots by prior-hit group	Diagnostic	Is proven talent safer? (hypothesis 4)
8	Regression coefficients with 95% intervals	Coefficient (dot-and-whisker) plot	Diagnostic	Which choices matter most once the others are held fixed?
9	Predicted outcome range for 3 example pitches	Fan / interval plot	Predictive	What should a studio expect from pitch A, B, C?
10	Slate risk-return frontier	Scatter with frontier line	Prescriptive	Best slate for a big studio vs a small one (hypothesis 5)

Charts 1, 2, 3, 8 and 10 are the core; if we run out of pages, 4 and 6 go first.

5. MODELS

Stage	Model	Target	Tool	Why this one
Diagnostic	Linear regression (OLS)	log gross multiple	statsmodels	Coefficients read as "% change in return", which is what the coefficient plot shows
Diagnostic	Logistic regression	flop (yes/no)	sklearn `LogisticRegression`	Same choices, but on risk instead of average return
Predictive	Gradient boosting with quantile loss	10th, 50th, 90th percentile of log multiple	sklearn `GradientBoostingRegressor(loss="quantile")`	Gives a range per film, not one number. Risk is the width of that range
Predictive baseline	Median multiple of the film's genre x budget tier cell	same	Polars	The model has to beat the simple rule a studio already uses
Prescriptive	Monte Carlo slate simulation	total slate profit	numpy	Turns predicted ranges into a decision. Brief's own example 4 uses Monte Carlo

Validation: train on films released up to 2011, test once on 2012-2016 (2019 where The Numbers covers it). A time split, because the studio always predicts the future from the past. Tuning inside the training set with 5-fold cross-validation and `GridSearchCV`, as in Lec9.

The prescriptive step: a studio has, say, $300M to spend in a year. It can fund one $150M tentpole plus three $50M films, or ten $30M horror and comedy films, and so on. For each candidate slate we draw 10,000 outcomes from the predicted ranges, convert gross to studio profit with the 2.5x rule, and record the expected profit and the chance the whole slate loses money. Plotting every slate gives a risk-return frontier, like a stock portfolio. A big studio that can absorb a bad year picks a different point on it than a small one that cannot.

6. LIMITATIONS TO STATE UP FRONT

1. Box office is not profit. Streaming, TV and home video revenue are missing, and so are real marketing budgets. The 2.5x rule stands in for both.
2. Actor and director pay is not public. Track record is a proxy for star power, not a salary.
3. Genre is a rough proxy for effects spending, not a measure of it.
4. Survivorship: the datasets favour films big enough to be tracked. Tiny films that vanished are underrepresented, which makes the low-budget tier look better than it is.
5. Associations, not causes. A big budget may lower flop risk because studios only give big budgets to projects they already trust.

7. NEXT STEPS

1. Build the joined table in a notebook (`movie-studio/movie_studio.ipynb`): clean, match, inflation-adjust, engineer track record and competition features.
2. Make charts 1 to 3 first and check that hypotheses 1 and 2 still hold on the joined data.
3. Diagnostic and predictive models.
4. Slate simulation.
5. Report (8 pages, no code) plus the notebook as HTML, zipped as `Project_Jupyter`, due **Fri 13 Nov 23:59**. Presentation (15 min, everyone speaks) in weeks 11-12, so charts 1-3 and the hypotheses need to exist before then.

Sources. The Numbers budgets via delafields.github.io/movie-budgets; TMDB 5000 (Kaggle mirror on Hugging Face); Lock5Data HollywoodMovies; FRED CPIAUCSL.